

# AI-Based Citizen Grievance Classification System (Precinct One)

**Problem Statement:** Develop an AI-based system that automatically analyzes citizen complaints and categorizes them into relevant departments to improve efficiency.

**Objective:** Transform unstructured complaints into structured outputs including category, priority, and department.

**Core Features:**

- Automated classification of complaints
- Multilingual support (English + Hindi)
- Priority detection
- Intelligent routing (simulated)

**System Architecture:**

User Input → Frontend → Backend API → AI Classification → Structured Output → Display

**Input Format:** { text: complaint description }

**Output Format:** Category, Subcategory, Priority, Department, Status

**Classification Logic:**

- Categories: Water, Electricity, Roads, Sanitation, Others
- Priority: High, Medium, Low based on urgency keywords
- Department mapping based on category

**Fallback Strategy:**

- If AI fails → keyword-based classification
- Ensures system reliability

**Edge Cases:**

- Empty input
- Mixed complaints
- Vague or spam input

**Scalability Plan:**

- Database integration
- Dashboard & analytics
- Real-time government API integration
- Custom ML model training

#### Conclusion:

Precinct One acts as an intelligent pre-processing engine improving classification accuracy, reducing manual effort, and enabling faster grievance handling.